XIANG LI

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EDUCATION

Peking University, Beijing, China Sep. 2018 – Jul. 2023
Ph.D. in Statistics, School of Mathematical Sciences
Advisor: Prof. Zhihua Zhang

Peking University, Beijing, China Sep. 2014 – Jul. 2018
B.S. in Statistics, School of Mathematical Sciences
B.A. in Economics, National School of Development

EXPERIENCE

Rutgers University, Piscataway, USA Starting Jan. 2027
Assistant Professor, Department of Statistics

University of Pennsylvania, Philadelphia, USA Aug. 2023 – Present
Postdoctoral Researcher. Working with Prof. Weijie Su and Prof. Qi Long

University of Toronto, Toronto, Canada Feb. 2022 – Feb. 2023
Visiting Ph.D. Student. Working with Prof. Qiang Sun

Face++ (Megvii), Beijing, China Jun. 2018 – Nov. 2018
Research Intern, Algorithm Group. Mentor: Dr. Shuchang Zhou

ToSimple, Beijing, China Mar. 2018 – May. 2018
Research Intern, Algorithm Group. Mentor: Dr. Naiyan Wang

SELECTED AWARDS

- IMS New Researcher Travel Award 2025
- Outstanding PhD Graduates, Peking University 2023
- President Scholarship for Excellent Ph.D Student, Peking University 2018, 2020-2021
- Outstanding reviewer in NeurIPS, top 8% 2021
- National Scholarship, China 2017, 2019
- Travel Award NeurIPS 2019, AAAI 2020
- Outstanding Graduates, School of Mathematical Sciences, Peking University 2018
- First prize in National Undergraduate Mathematical Modeling Contest, China 2016

RESEARCH INTERESTS

My research lies at the intersection of statistics, machine learning, and optimization. I develop statistical methodology and theory for modern data-scientific problems, with emphasis on inference, uncertainty quantification, robustness, privacy, and scalable learning algorithms. A major part of my recent work studies generative AI and large language models from a statistical perspective, including provenance, evaluation, uncertainty, and reliability. More broadly, my research spans the following directions:

- **Statistical Foundations of Modern Machine Learning**

- **Generative AI, Provenance, and Reliability.** Developing statistical methods for analyzing and verifying outputs from generative models, including watermarking schemes [J4, C3, J5, C4], detection procedures [J1, J2, C5, S3], and estimation methods [J3, S1] for provenance, accountability, and reliability.

- **Evaluation and Uncertainty Quantification.** Modeling complex learning systems as statistical black boxes and developing tools to study input–output behavior, with applications to model evaluation [S4, C7], uncertainty quantification [S5], scaling laws, privacy-preserving analysis, and distributional shift.

- **Reliable Statistical Machine Learning**
  - **Learning under Challenging Conditions.** Developing efficient algorithms for nonstandard settings such as heavy-tailed noise [J10], corruption [C8], high-dimensionality, and data heterogeneity, and analyzing their statistical and convergence properties [C15, C10].
  - **Inference in Adaptive and Dynamic Environments.** Designing statistical methods for streaming data [J6, J8] and adaptive decision-making [C9, C14, C16], with the goal of ensuring valid inference in dynamic, dependent, or data-adaptive environments [C2, T1, J7, S7].

- **Federated Learning, Privacy, and Distributed Statistics**
  - **Communication-Efficient Learning.** Developing communication-efficient algorithms [J11, C13, T3] and analyzing their convergence properties under heterogeneous data settings [C1, C11, T2].
  - **Trustworthy Distributed Learning.** Studying statistical and algorithmic aspects of trustworthy federated learning, including differential privacy [C6, S6], personalization [J9, C12], and robust distributed inference.

PUBLICATIONS

* indicates equal contribution. ** indicates alphabetical order.

Journal articles (Published or under revision)

[J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Annals of Statistics, 53 (1): 322-351, Feb. 2025
Selected for presentation at the Annals of Statistics Invited Paper Session, JSM 2025.

[J2] **Robust Detection of Watermarks in Large Language Models Under Human Edits** [pdf] [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Journal of the Royal Statistical Society Series B: Statistical Methodology, 88 (2): 491–515, Apr. 2026
Received the 2025 IMS New Researcher Travel Award
Selected for presentation at JRSSb Editors Invited Session at the RSS Conference 2026

[J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]
Xiang Li, Garrett G. Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su
To appear in *Journal of the American Statistical Association*, 2026.

[J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [pdf] [arxiv]
Yangxinyu Xie, **Xiang Li**, Tanwi Mallick, Weijie J. Su, Ruixun Zhang
Journal of the American Statistical Association, 1–21, 2025

[J5] **Robust Spectral Watermark for Synthetic Tabular Data** [pdf]
Yizhou Zhao, **Xiang Li**, Peter Song, Qi Long, Weijie J. Su

Statistical Learning and Data Science

[J6] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control** [pdf]
Xiang Li*, Jiadong Liang*, Xinyun Chen, Zhihua Zhang
Operation Research, Jan. 2026

[J7] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [pdf] [arxiv]
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang
Mathematics of Operations Research, published online Aug. 2026

[J8] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [pdf] [arxiv]
Yuze Han**, **Xiang Li****, Zhihua Zhang**
Journal of Machine Learning Research, 27 (112): 1–69, 2026.

[J9] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness** [pdf]
Yuze Han**, **Xiang Li****, Shiyun Lin**, Zhihua Zhang**
Journal of Machine Learning Research, 25 (380): 1-88, Dec. 2024

[J10] **Variance-Aware Decision Making with Linear Function Approximation under Heavy-Tailed Rewards** [pdf] [arxiv]
Xiang Li, Qiang Sun
Transactions on Machine Learning Research, Apr. 2024
Invited to present at ICLR 2025; ranked among the Top 100 papers accepted by TMLR that year.

[J11] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation** [pdf] [arxiv]
Xiao Guo*, **Xiang Li***, Xiangyu Chang, Shusen Wang, Zhihua Zhang
Machine Learning, 113 (11): 8427-8458, Sep. 2024

[J12] **Discussion of "Balanced and Robust Randomized Treatment Assignments: The Finite Selection Model for the Health Insurance Experiment and Beyond"** [pdf]
Xiang Li, Yangxinyu Xie, Weng-Kee Wong
Journal of the Royal Statistical Society Series A: Statistics in Society, 2026 (Discussion article)

Conference papers

[C1] **On the Convergence of FedAvg on Non-IID Data** [pdf]
Xiang Li*, Kaixuan Huang*, Wenhao Yang*, Shusen Wang, Zhihua Zhang
International Conference on Learning Representations (ICLR) 2020 (Oral presentation, Google citation 3300+)

[C2] **Statistical Estimation and Online Inference via Local SGD** [pdf]
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang
Conference on Learning Theory (COLT) 2022

[C3] **Improving the Trade-off Between Watermark Strength and Speculative Sampling Efficiency for Language Models** [pdf]
Weiqing He*, **Xiang Li***, Li Shen, Weijie J. Su, Qi Long
International Conference on Learning Representations (ICLR) 2026

[C4] **Selective Disclosure Watermarking for Large Language Models** [arxiv] [pdf]
Xuyang Chen, **Xiang Li**, Yangxinyu Xie, Qi Long
International Conference on Machine Learning (ICML) 2026

[C5] **On the Empirical Power of Goodness-of-Fit Tests in Watermark Detection** [arxiv]
Weiqing He* and **Xiang Li***, Tianqi Shang, Li Shen, Weijie J. Su, Qi Long
Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)

[C6] **Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via f-Differential Privacy** [arxiv]
Xiang Li*, Buxin Su*, Chendi Wang*, Qi Long, Weijie J. Su
Neural Information Processing Systems (NeurIPS) 2025 (Spotlight)

[C7] **UCS: Estimating Unseen Coverage for Improved In-Context Learning** [pdf]
Jiayi Xin, **Xiang Li**, Evan Qiang, Weiqing He, Tianqi Shang, Weijie J. Su, Qi Long
Findings of the Association for Computational Linguistics (ACL) 2026

[C8] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards** [pdf]
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun
Conference on Uncertainty in Artificial Intelligence (UAI) 2025

[C9] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [pdf]
Xiang Li, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C10] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [pdf]
Hengchao Chen, **Xiang Li**, Qiang Sun
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C11] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [pdf]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022 (Spotlight)

[C12] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [pdf]
Shiyun Lin*, Yuze Han*, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022

[C13] **Communication-Efficient Distributed SVD via Local Power Iterations** [pdf]
Xiang Li, Shusen Wang, Kun Chen, Zhihua Zhang
International Conference on Machine Learning (ICML) 2021

[C14] **Finding Near Optimal Policies via Reducive Regularization in Markov Decision Processes** [pdf]
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang
Workshop on Reinforcement Learning Theory, ICML 2021

[C15] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [pdf]
Xiang Li, Shusen Wang, Zhihua Zhang
AAAI Conference on Artificial Intelligence (AAAI) 2020

[C16] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [pdf]
Wenhao Yang*, **Xiang Li***, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2019

Preprints and submitted manuscripts

[S1] **Optimal Watermark Localization in Mixed-Source Large Language Model Texts** [arxiv]
Jose H. Blanchet**, T. Tony Cai**, **Xiang Li****, Hao Liu**, Qi Long**, Weijie J. Su**

[S2] **Steer-to-Detect: Probing Hidden Representations for Detection of LLM-Generated Texts** [arxiv]
Luxu Liang, **Xiang Li**

[S3] **Optimal Detection for Language Watermarks with Pseudorandom Collision** [arxiv]
T. Tony Cai**, **Xiang Li****, Qi Long**, Weijie J. Su**, Garrett G. Wen**
Submitted to *Annals of Statistics*

[S4] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [arxiv]
Xiang Li, Jiayi Xin, Qi Long, Weijie J. Su
Submitted to *Proceedings of the National Academy of Sciences (PNAS)*

[S5] **Paraphrase-Robust Conformal Prediction for Reliable LLM Uncertainty Quantification**
Jiayi Xin, Evan Qiang, **Xiang Li**, Weijie J. Su, Qi Long
Submitted to *International Conference on Machine Learning (ICML) 2026*

[S6] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [arxiv]
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma
Submitted to *Journal of Computational and Graphical Statistics*

[S7] **Uncertainty Quantification of Data Shapley via Statistical Inference** [arxiv]
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang
Submitted to *Machine Learning*

Technical reports

[T1] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** [arxiv] [code]
Xiang Li, Jiadong Liang, Zhihua Zhang

[T2] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** [arxiv]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang

[T3] **Communication Efficient Decentralized Training with Multiple Local Updates** [arxiv]
Xiang Li, Wenhao Yang, Shusen Wang, Zhihua Zhang

GRANTS & FUNDING

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie J. Su & Prof. Qi Long (2024)
  - Assisted in proposal writing, structuring key research points on watermarking.

PRESENTATIONS

- *Optimal Detection for Language Watermarks with Pseudorandom Collision*
  - Mar. 2026, ENAR Spring Meeting, Indianapolis.
  - Jun. 2026, ICSA Applied Statistics Symposium, Arlington.
- *Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?*
  - Sep. 2025, AI Research Mixer 2025, UPenn.
- *Robust Watermark Detection and Efficient Proportion Estimation for Human-Edited Watermarked Text*
  - Jun. 2025, ICSA Applied Statistics Symposium, University of Connecticut.
  - Aug. 2025, Joint Statistical Meetings, Nashville, Tennessee.

- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
  - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*
  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

TEACHING EXPERIENCES

Instructor

- Short course (*Statistical Inference in Large Language Models*) at the 2025 ICSA Applied Statistics Symposium [slides] Summer 2025

Invited guest lecturer

- STAT 9917: *Statistical Topics in Large Language Models*, UPenn Spring 2025

Teaching assistant

- *Linear Algebra*, Peking University Spring 2021
- *Reinforcement Learning: Theory and Algorithms*, Peking University Fall 2019

ACADEMIC SERVICE

Conference review

- NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.

Journal review

- Journal of Machine Learning Research
- Transactions on Machine Learning Research
- Journal of the American Statistical Association
- Information and Inference: A Journal of the IMA
- Annals of Applied Probability
- Operational Research
- IEEE Transactions on Automatic Control
- IEEE Open Journal of Signal Processing
- IEEE Journal on Selected Areas in Communications
- Transactions on Parallel and Distributed Systems
- World Wide Web

Event organization

- Aug. 2024, Session Chair for "Theoretical and Algorithmic Developments in Federated Learning" at the Modeling and Optimization: Theory and Applications (MOPTA) conference.